

ADMINISTRATIVE CONTROL, LIVE ART BIDDING, 3D MODELLING

AEMS - Scenario 2

Group 12 - Aditi Ghosh, Brian Daza, Abrar Makki

For scenario 2 we will mainly focus on the administrative perspective (system user/system admin) of our application AEMS apart from extending existing features. So, we will be expanding our system's capabilities by incorporating features like:

- Advanced Admin management
- Live art bidding i.e. the auction module
- 3D modeling of the artworks
- Real time analytics and reporting

Part 1: Admin role and management

Upon logging in as the system admin, the system verifies the credentials and it redirects the user to the main admin dashboard. The admin can have full operational control over users, content, and platform settings. They will have access to management options/controls like

Managing artwork

On selecting the button option for “**Manage Artworks**”, the admin will be able to view the complete list of artworks available in the system. The list will display basic details such as artwork title, artist name, category, price, and current status (Available, Sold, In Auction). The admin can choose to **Add New Artwork** option, where the system asks for details including artwork title, artist name, description, dimensions, medium, and price. The admin can also upload high-resolution images and optionally attach a 3D model file for immersive viewing. They can even modify the existing artworks like **Edit artwork information, Update price, Change status (Available / Sold / In Auction), Delete artwork, Enable auction mode.**

After making the necessary changes, the system saves the updated information and reflects the changes on the customer side of the platform.

Managing exhibitions

On selecting “**Manage Exhibitions**”, the admin can create and manage upcoming or ongoing exhibitions. The system allows the admin to enter details such as: **Exhibition title, Venue, Location, Date and time, Participating artists, Featured artworks**

The admin can edit or remove exhibitions as required. Once finalized, the exhibition details are published on the customer dashboard under “View Upcoming Events.”

Manage auctions (Live art bidding)

On selecting the “**Manage Auctions**” option, the admin can initiate or configure live bidding for selected artworks. The admin selects an artwork and sets: **Starting bid amount, Minimum bid**

increment, Auction start time, Auction end time. Once the auction is activated, the artwork will become visible to customers under the bidding section which we will implement as an extended functionality under customer side interface. During the auction, the admin will be able to monitor live bids, **view the current highest bidder, and track bid history**. After the auction ends, the system will automatically declare the highest bidder as the winner and generate a payment request. The artwork status will update accordingly. The admin can also if required manually pause, cancel, or conclude the auction.

View reports and analytics

On selecting “**Reports & Analytics**”, the admin can view summarized performance data of the system. The dashboard displays: **Total ticket sales, Total art sales revenue, Active auctions, Most viewed artworks, Most bid artworks, Customer purchase trends**

The admin will be able to generate reports based on selected filters and export them for further analysis. This allows the admin to make data-driven decisions regarding exhibitions, pricing, and marketing strategies.

Part 2: Extending customer functionality

3D modeling of artworks

3D renders of the art pieces will show up when a customer clicks on a piece they are interested in. Several actions the user can do within the art renderer is to rotate it, zoom in to see the fine details, and get a better sense of the quality of the piece being sold, allowing customers to better assess quality, texture, and craftsmanship before purchasing. This enhances user engagement and provides an immersive exhibition experience. If a 3D model is not available, the system defaults to standard image viewing.

Additional customer features to extend

Now, the buy art page will include multiple new ways to buy, one of which will be from actions. The server will try to search for best prices online for a given art piece and determine whether that piece is being auctioned or sold individually. Auctioned art pieces will be accompanied by a warning for the user to understand that they must place a bid and that their bid is not guaranteed to go through for the full duration of the auction. Auctioned pieces will have a timer listed onto them that will show how long the customer is allowed to place a bid before it is sold off. The bids are made public, but users can request their bid to be made anonymous. These bids will be kept track of in a table next to the art pieces selected.

Class Diagram:

<https://docs.google.com/document/d/1yViv4vrryJ2XpSDjxoNGQJHBsz7lCE1SM69wcvysO5c/e/dit?usp=sharing>